MIL-C-52712A(ME)
16 August 1982
SUPERSEDING
MIL-T-52712(ME)
22 July 1971

MILITARY SPECIFICATION

CONVEYOR, BELT, PORTABLE, ELECTRIC, CASTER MOUNTED

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a portable, electric motor driven conveyor equipped with a flat belt and caster mounts.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel and Seals.
TT-P-664	- Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
TT-V-121	- Varnish, Spar, Water Resisting.
PPP-B-601	- Boxes, Wood, Nailed and Lock-Corner.
PPP-P-40	- Packaging and Packing of Hand Tools.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3910

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MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-T-22085	- Tape, Pressure-Sensitive, Adhesive, Preservation and Sealing.
MIL-B-26195	- Box, Wood-cleated, Skidded, Load-bearing Base.

STANDARDS

FEDERAL

FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

Safety Standards for Conveyors and Related Equipment (ANSI B20.1).

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

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CASTER AND FLOOR TRUCK MANUFACTURER'S ASSOCIATION (CFTMA)

Industrial Wheel Standard, Manually Operated or Power Drawn at No More Than 3 MPH Wheel Types.

(Applicaton for copies should be addressed to the Caster and Floor Truck Manufacturer's Association, 1717 Howard Street, Evanston, IL 60202.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The portable conveyor is a continuous belt type with steel slider plates, electric-motor-driven item used to move materials horizontally or up an incline.

3.1.1 Dimensions. The conveyor shall have a nominal length of 20 feet with a belt width of 18 inches and shall weigh not more than 500 pounds.

3.1.2 Capacity and speed. The conveyor shall have a capacity of not less than 45 pounds per linear foot when the conveyor is level and not less than 300 pounds total when the conveyor is operated at an angle of 30 degrees. Belt speed shall be not less than 100 FPM (feet per minute).

3.2 First article. The contractor shall furnish one or more conveyors as specified (see 6.2), for examination and demonstration within the time frame specified (see 6.2), to prove that the conveyors being furnished comply with the requirements of this specification. Examination and demonstration shall be as specified in Section 4 and shall be subject to the surveillance and approval by the government.

3.3 Safety. The conveyor shall incorporate all safety features and design characteristics of ANSI Standard B20.1. An anti-runway brake or back to stop device shall be provided to prevent personnel injury. Emergency stop devices shall be provided at all operator locations. All rotating, reciprocating, or moving parts which are of such nature or so located as to be hazardous to operating or maintenance personnel shall be fully enclosed or guarded. All exposed parts which are subject to high operating temperatures or which are energized electrically shall be insulated, fully enclosed or guarded.

3.4 Frame. The conveyor frame shall be aluminum. Tubular handles shall be provided near the head and tail end to assist in moving and positioning the conveyor.

3.4.1 Mounting. Each conveyor section shall be mounted on four casters. The casters shall be belted to the conveyor frame, two near the head end and two near the tail end.

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3.4.2 Casters. Casters shall be heavy duty, swivel plate, scaffold type with foot operated brakes. They shall be equipped with ball bearings and shall accommodate the specified wheel. Swivels shall be provided with grease fittings.

3.4.3 Caster wheels. Caster wheels shall be rubber tired and a minimum of 6 inches in diameter by 2 inches rim width. They shall conform to the operational load ratings of the CFTMA standard for industrial wheels. The wheels shall be mounted on ball or roller bearings.

3.5 Pulleys, idlers and rollers.

3.5.1 Pulleys. Head pulley, tail pulley and drive pulley shall be bearing mounted. Bearings shall be the lubricated for life type. A screw type adjuster with positive locking means shall be provided on the head and tail pulleys.

3.5.2 Idler rollers. Idler rollers shall be bearing mounted. Bearings shall be the lubricated for life type.

3.5.3 Take-up rollers. Take-up rollers shall be bearing mounted. Bearings shall be lubricated for life type. A screw type adjuster with positive locking means shall be provided.

3.5.4 Conveyor belt. A rubber, rough top type, conveyor belt shall be provided. It shall be an 18 inch wide continuous belt with alligator type lacing. Means for adjusting belt tracking shall be provided.

3.6 Drive train. The drive train shall be gear or roller type chain and sprocket. When chain drive is furnished, a spring loaded chain tensioner and idler sprocket shall be provided to maintain constant chain tension.

3.7 Electrical equipment. The conveyor shall be equipped with an electric motor, a system of electric wiring and controls and a power supply cable. It shall have a magnetic start and a push button start-stop switch. The conveyor shall be wired for 230 VAC, 60 hertz, single phase power.

3.7.1 Electric motor. The conveyor belt shall be driven by an electric motor rated for 1 horsepower minimum, 115/230 volts, 60 hertz, single phase a.c.

3.7.2 Starter. Starter shall be of the magnetic reversing type with manual reset overload relay.

3.7.3 Controls. Push button start and stop controls shall be provided near the head end and the tail end of the conveyor. Both stop push buttons shall have means for locking in the stop position. In addition, a reverse-stop-forward switch shall be mounted near the tail end.

3.7.4 Power cable. A 40 foot, No. 12 AWG, three wire, type s.o. electrical cord with a three prong male connector shall be provided and shall be connected to the starter. A mating three wire female receptacle compatible with the male connector shall be supplied with each conveyor.

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3.8 Lubrication. Lubrication means shall be provided for all moving parts that require lubrication. The types of lubricants shall be selected from, and be in accordance with, SAE J754. All initial fills of lubricants shall be limited to the types approved by the component manufacturers. Grease lube fittings shall conform to SAE J534.

3.9 Identification marking. Each conveyor section shall be identified with an identification plate conforming to MIL-P-514, Type I, Style 1, Composition C, of Type I, Grade A, Class 1 material which shall be located in a visible and permanent location. The identifying data shall be stamped in letters in accordance with MIL-P-514. The plate shall be securely attached to the conveyor with screws, bolts. or rivets, and shall be furnished, filled out and mounted by the contractor.

3.10 Treatment and painting. The parts of the conveyor normally painted shall be cleaned, pretreated, and painted in accordance with MIL-T-704, Type A or an equivalent commercial procedure. The final color shall be No. 13538 in accordance with FED-STD-595.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections, evaluations and demonstration. Inspections, evaluations, and demonstrations shall be classified as follows:

a. First article evaluation and demonstration (see 4.3).
b. Production run-in (see 4.5).
c. Inspection of packaging (see 4.7).

4.3 First article evaluation and demonstration. Prior to delivery the manufacturer shall make one or more conveyors available, selected at random from conveyors being produced from production tooling, for first article evaluation and demonstration. The conveyor and its components shall be compared with this document, published literature, calculations and test data to verify compliance with this specification. The conveyor shall then be subjected to a demonstration as specified herein. Permanent deformation, overheating, malfunction, failure to meet performance requirements, failure to complete any demonstration, or leakage of lubricants shall be cause for rejection of the conveyor. If such deficiencies do arise, they shall be corrected and the affected demonstration shall be repeated as required.

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4.4 Demonstration.

4.4.1 Demonstration conditions. Unless otherwise specified herein, demonstration shall be conducted at the ambient temperature and climatic conditions existing at the place of demonstration. Only that maintenance established by the contractor and submitted as a maintenance schedule prior to commencement of the demonstration shall be performed during the demonstration.

4.4.2 Conveyor operation. The conveyor shall be operated continuously at maximum capacity and speed for a period of 2 hours. The load shall be transported up a slope of not less than 30 degrees. Evidence of overheating, excessive wear, malfunction, inability to complete the operation or failure of any component shall be cause for rejection.

4.5 Production run-in. Each conveyor shall be operated at the capacity and speed specified in 3.1.2 for a period of 30 minutes. It shall also be operated in an unloaded condition for 30 minutes. Inability to complete the run-in, overheating, failure of any component, malfunction of any control, or evidence of leakage of fluids shall be cause of rejection of the conveyor by the Government until such deficiencies are corrected and the run-in successfully performed.

4.6 First article demonstration report. The contractor shall prepare and submit the first article demonstration report to the contracting officer for approval within the time frame specified (see 6.2). The report shall include a detailed description of the examination and demonstration performed. All deficiencies encountered and actions taken to correct them shall be included.

4.7 Inspections of packaging.

4.7.1 Preproductions pack inspection.

4.7.1.1 Examination. Examine the preproduction pack for the defects specified in 4.7.2.3. Presence of one or more defects shall be cause for rejection.

4.7.2 Quality conformance inspection of pack.

4.7.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.7.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.7.2.3 Examination. Samples selected in accordance with 4.7.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

101. Materials, methods, and containers not as specified for level A. Each incorrect material, method or container shall be considered one defect.
102. Disassembly not as specified for level A.
103. Unprotected surfaces not coated with preservative as specified for level A.

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104. Belt fasteners not coated with varnish as specified for level A.
105. Conveyor belt not rolled and roll secured as specified for level A.
106. Electrical components not sealed as specified for level A.
107. Cables not coiled and secured as specified for level A.
108. Drive chains not preserved as specified for level A.
109. Gears not preserved as specified for level A.
110. Repair parts not preserved in accordance with the referenced document for level A.
111. Maintenance tools not preserved in accordance with the referenced document for level A.
112. Consolidation not as specified for level A.
113. Blocking, bracing and anchoring not in accordance with referenced document for level A.
114. Marking missing, illegible, incorrect, or incomplete for level A or commercial.

5. PACKAGING

5.1 Preproduction pack. The contractor shall furnish a preproduction pack for examination within the time frame specified (see 6.2), to prove, prior to starting production packaging that the applied preservation, packing and marking comply with the packaging requirements of this specification. Examination shall be as specified in Section 4, and shall be subject to surveillance and approval by the Government (see 6.3). The preproduction pack may be accomplished utilizing either the preproduction model conveyor or a production conveyor. If the preproduction model conveyor is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government when requested by the Government to facilitate comparison between the preproduction model conveyor and the production conveyor.

5.2 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Disassembly. The conveyor belt shall be removed. Further disassembly shall be the minimum necessary to safeguard parts known to be subject to damage or loss in accordance with the contractor's recommendations. Bolts, nuts, pins, and washers involved shall be placed in one of the mating parts and secured to prevent loss.

5.2.1.2 Preservatives. Preservatives specified herein shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.2.1.3 Unprotected surfaces. Unpainted exterior metal surfaces of component parts requiring the application of a contact preservative in accordance with MIL-P-116, such as bolts, nuts, pins, washers, shafting, rollers, and any other unprotected metal surfaces not specified herein shall be coated with type P-1 preservative. The preservative shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

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5.2.1.4 Conveyor belt. Unprotected belt fasteners shall be coated with varnish conforming to TT-V-121. The belt shall be compactly rolled and the roll shall be secured with two straps conforming to QQ-S-781, Class 1, Type I or IV, Finish B, size as applicable. The strapping shall be placed circumferentially on the roll and the straps tensioned so as not to damage the belt.

5.2.1.5 Electrical components. Openings into the motor, controls, junction box, or receptacles shall be sealed with tape conforming to MIL-T-22085, Type II. The power cable shall be coiled and the coils secured with tape specified herein. Cable connected to components shall be secured to the frame of the conveyor with the tape specified herein. Loose cables shall be consolidated as specified in 5.2.1.11.

5.2.1.6 Drive chains. Exposed drive chains shall be coated with enough type P-3 or P-9 preservative to assure penetration of the preservative to the inner surfaces of the rollers, pins, and bushings. The excess preservative shall be drained and the chain recoated with type P-1 preservative. Removed chains shall be coiled and the coil secured with soft annealed wire and wrapped with barrier material conforming to MIL-B-121, Type I, Grade A, Class 2, secured with tape conforming to MIL-T-22085, Type II. Sprockets and other unprotected surfaces of the drives shall be coated with type P-1 preservative.

5.2.1.7 Gears.

5.2.1.7.1 Exposed gears. Exposed gears shall be coated with type P-1 preservative or with primer conforming to TT-P-664.

5.2.1.7.2 Enclosed gears. Gear housings shall be filled to the operating level with the operational lubricant and the gears actuated to assure coating to all interior parts and surfaces with the lubricant.

5.2.1.8 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve repair parts.

5.2.1.9 Maintenance tools. Maintenance tools shall be preserved in accordance with the level A, preservation and packaging requirements of PPP-P-40.

5.2.1.10 Technical publications. Maintenance tools shall be preserved in accordance with MIL-P-116, Method IC-1 or IC-3.

5.2.1.11 Consolidation. Preserved and unpreserved components not attached to the conveyor shall be consolidated together in a close-fitting box conforming to PPP-B-601, Overseas type, style optional. Strapping shall not be required.

5.2.2 Commercial. Each complete conveyor shall be preserved in accordance with MIL-STD-1188.

5.3 Packing. Packing shall be level A or commercial as specified (see 6.2).

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5.3.1 Level A. Each complete conveyor shall be packed in a box conforming to MIL-B-26195, Type II, Style C, Class 1 or 2, with plywood superstructure and rubbing strips. The length requirements of the box shall be waived. The contents shall be blocked, braced and anchored in accordance with MIL-STD-1186.

5.3.2 Commercial. Each complete conveyor shall be packed in accordance with MIL-STD-1188.

5.4 Marking.

5.4.1 Military. Marking for military level A shall be in accordance with MIL-STD-129.

5.4.2. Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The electric power-belt conveyor is intended for use in loading and unloading cartons, boxes, or bags in trucks and freight cars, and to move the packages from one level to another or in a continuous level plane, and help in stacking and piling the containers in warehouses.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
c. Number of conveyors to be furnished. Time frame required for submission of first article coveyor (see 3.2).
d. Time frame for article demonstration report (see 4.6).
e. Time frame required for submission of the preproduction pack.
f. Degree of preservation and degree of packing required (see 5.2 and 5.3)

6.3 Preproduction pack. Any change or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, pack, and mark the conveyors in conformance to this specification.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-52712A(ME) Conveyor, Belt, Portable, Electric, Caster Mounted

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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